AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 22 · STUDENT LESSON

Pooling and spatial information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Declare the window. Compute both summaries. Find the lost detail.

Audit max and mean pooling with complete collision,negative-value and boundary counterexamples.

Keep one habit: Four-region arithmetic and diagram → pictured within-window swap → explicit boundary-movement head → runnable probes → independent negative-valued map.

MAIN LESSON · SECTION 1

Which spatial detail survives a local summary?

> Compute the summary and inspect the collisions and boundaries it creates.

Learning goals

I can compute max and **mean pooling** under an explicit window rule.

I can use collisions to explain lost **spatial information**.

I can test boundary and movement claims with revealing examples.

Retrieve local windows and their contract

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- For a 3 by 3 **input** and a 2 by 2 window with stride 2 and no padding, how many valid window positions exist?

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

By the end, I can explain pooling and **spatial information** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Compute max and **mean pooling** under an explicit window rule

Use collisions to explain lost **spatial information**

Test boundary and movement claims with revealing examples

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Summarize four complete non-overlapping windows

> Max and mean are separate fixed operations on the same input.

Complete feature map and window rule

A fictional single-channel feature map has rows [1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3] and [2, 4, 2, 7]. Use a 2 by 2 window, stride 2, no padding and only complete windows. Coordinates are zero-based.

The four starts are [0, 0], [0, 2], [2, 0] and [2, 2].

Max pooling keeps the largest value in each window. **Mean pooling** adds the four values and divides by 4. Neither operation learns weights in this lesson.

These are fixed summaries of the incoming feature map, unlike the trainable kernel parameters introduced last week.

Every window and both summaries

Window start	Values	Maximum	Mean
[0, 0]	1, 4, 3, 2	4	2.5
[0, 2]	2, 0, 5, 1	5	2
[2, 0]	0, 6, 2, 4	6	3
[2, 2]	1, 3, 2, 7	7	3.25

Output shape and retained coarse arrangement

The max **output** is [[4, 5], [6, 7]], and the mean **output** is [[2.5, 2], [3, 3.25]]. Both have shape 2 by 2. Sixteen values become four, but the four **output**s still preserve the coarse arrangement of the four regions.

Name given and missing facts

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Trace every region into both alternative summaries

> Keep the window rule and lost-detail claim at the same scale.

Complete function scope and boundary assumptions

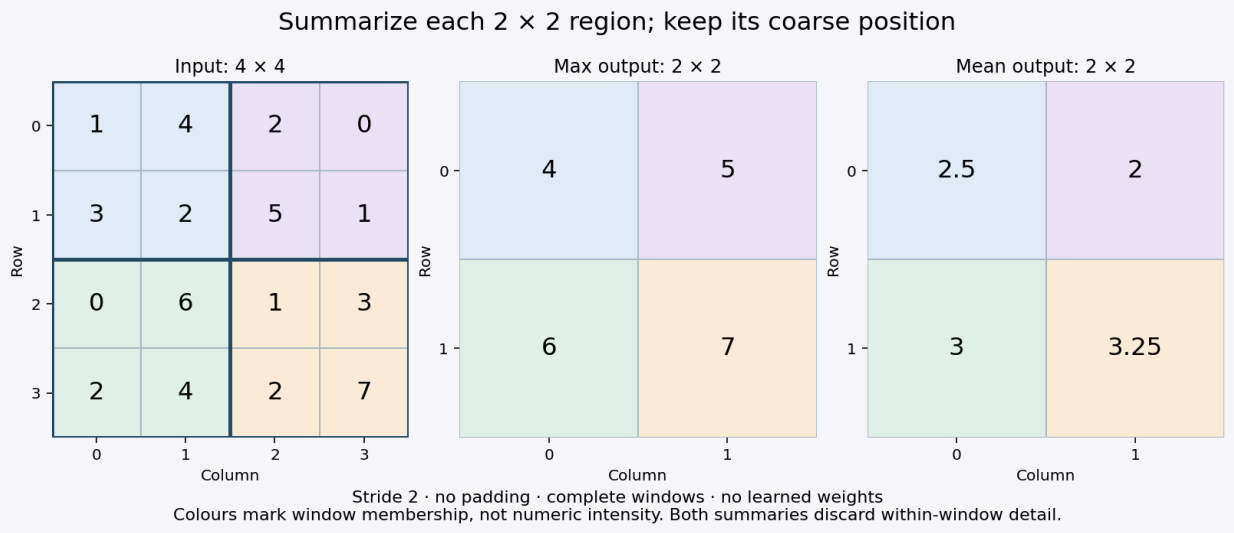
This code assumes a nonempty rectangular numeric map at least 2 by 2. It uses only complete 2 by 2 windows, stride 2 and no padding.

It returns two separate summaries of the same map; it does not apply one pooling operation after the other.

For this complete-window rule the **output** shape is floor((H - 2) / 2) + 1 by floor((W - 2) / 2) + 1. A 3 by 3 map produces just one **output**; its last row and last column are not read.

Do not silently invent a partial-window rule or pad missing positions.

Labelled four-region pooling diagram



All sums, denominators and maxima

Window start	Values in row-major order	Sum /4	Maximum
(0, 0)	[1, 4, 3, 2]	10/4=2.5	4
(0, 2)	[2, 0, 5, 1]	8/4=2	5

Window start	Values in row-major order	Sum /4	Maximum
(2, 0)	[0, 6, 2, 4]	12/4=3	6
(2, 2)	[1, 3, 2, 7]	13/4=3.25	7

Separate summaries and information contract

The two output maps preserve which coarse region produced each scalar. They do not preserve the four original positions and values within that region. pool 2 returns maxima and means separately from the original map. It requires a nonempty rectangular numeric input and checks that at least one 2×2 window fits. Only complete windows at stride 2 are read; with a3×3 input, the last row and column are excluded.

This boundary rule is part of the operation, not a discovered property of the data.

Complete runnable max/mean pooling

```
def pool2(image):
    h, w = len(image), len(image[0])
    if h < 2 or w < 2:
        raise ValueError("Need a complete 2 by 2 window")
    maxima, means = [], []
    for r in range(0, h - 1, 2):
        max_row, mean_row = [], []
        for c in range(0, w - 1, 2):
            values = [image[r+i][c+j]
                      for i in range(2) for j in range(2)]
            max_row.append(max(values))
            mean_row.append(sum(values) / 4)
        maxima.append(max_row)
        means.append(mean_row)
    return maxima, means

x = [[1, 4, 2, 0], [3, 2, 5, 1],
     [0, 6, 1, 3], [2, 4, 2, 7]]
assert pool2(x) == ([[4, 5], [6, 7]],
                    [[2.5, 2], [3, 3.25]])
```

Connect each coloured region to its output position

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Rearrange values within one region before checking its summaries

> A collision gives a concrete limit on what the pooled value can tell us.

Current position-loss task

- Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured within-window swap

- Change: Swap 1 and 4 within the left window.

▼ Check after predicting

Prediction: Both pooled maps remain unchanged.

Reason: Max and mean ignore the positions of values inside the same window.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See a changed arrangement with an unchanged local summary

> Keep within-window invariance separate from a whole-image movement claim.

Complete pictured map and distinct scope

Pool a small map and show the detail it cannot retain.

Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses only the top two rows of the main map:[[1, 4, 2, 0], [3, 2, 5, 1]]. Its two complete windows give max[[4, 5]]and mean[[2.5, 2]], shape 1×2. Swapping1 and4 inside the left window changes positions but not its maximum 4or mean 2.5. That pair cannot be reconstructed from either unchanged scalar alone. The main 4×4 case has four coarse regions; do not copy its 2×2 output shape into this separate picture.

Trace both windows and summaries



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Left window	Values 1, 4, 3, 2: max 4, mean 2.5.
2. Right window	Values 2, 0, 5, 1: max 5, mean 2.
3. Compare summaries	Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

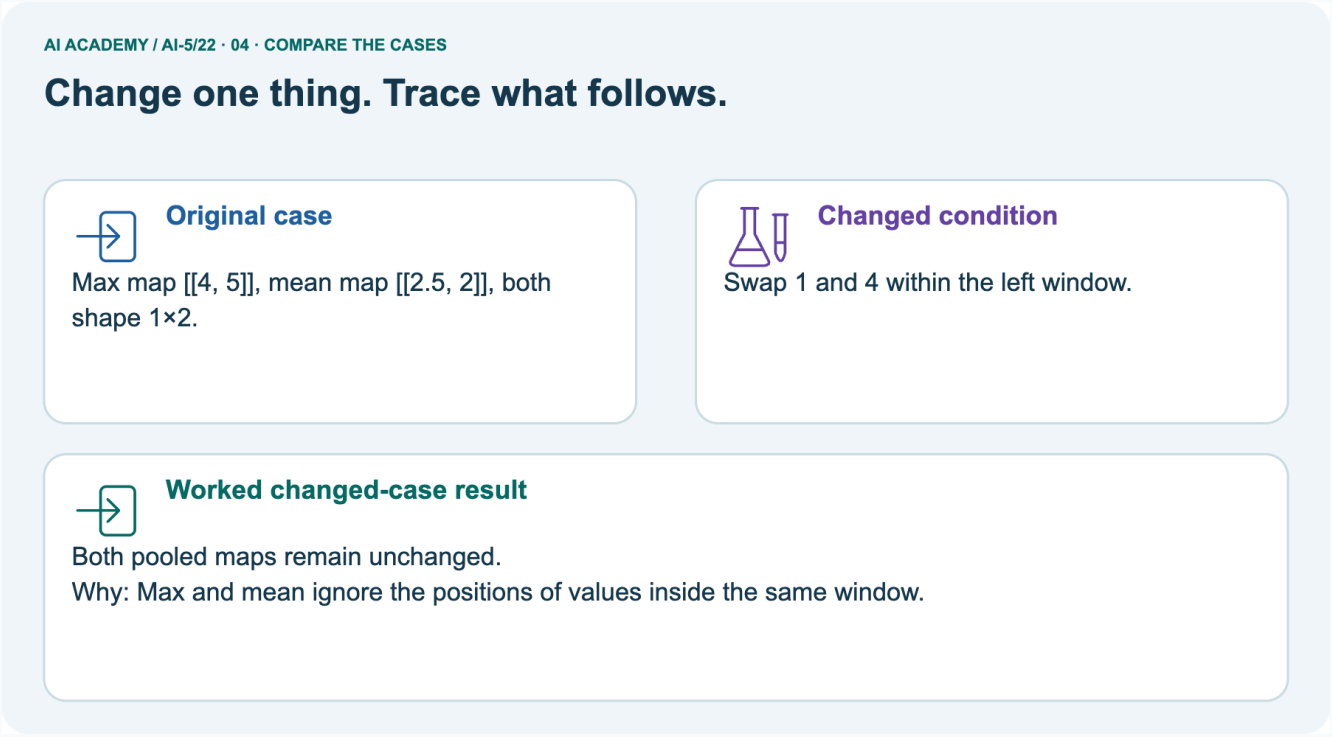
Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Predict the within-window swap

- Change one thing: Swap 1 and 4 within the left window. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain location loss and task limits

- Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

- Read the labelled picture: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the two bottom-position collision windows

A maximum does not identify its location

INPUT WINDOW ROWS	MAXIMUM	MEAN
[4 0] and [0 0]	4	1
[0 4] and [0 0]	4	1
[4 4] and [4 4]	4	4

The first two windows differ in the position of their nonzero value yet have the same maximum and mean. The third has the same maximum but a different mean. Each summary retains some distinctions and discards others.

Compare windows [[0, 0], [6, 0]] and [[0, 0], [0, 6]]. Compute both summaries. Can a deterministic head receiving only either scalar distinguish bottom-left from bottom-right?

▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Pooling and spatial information

The facts we will use

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Two adjacent pooling windows

	left 0	left 1	right 0	right 1
row 0	1	4	2	0
row 1	3	2	5	1

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/22

Pooling and spatial information

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Left window

Values 1, 4, 3, 2: max 4, mean 2.5.



2. Right window

Values 2, 0, 5, 1: max 5, mean 2.



3. Compare summaries

Max map [[4, 5]], mean map [[2.5, 2]], both shape 1 x 2.

CHANGE ONE THING

Swap 1 and 4 within the left window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Test movement across a boundary and all-negative windows

> Pooling preserves some rearrangements,not every spatial change.

Complete boundary,negative-value and max-position limits

Within one isolated window, rearranging its values preserves both its maximum and mean. But moving a value across a window boundary can change the **output** map. In a 2 by 4 map of zeros, put a single 8 at row 0 column 1.

Max pooling gives [[8, 0]]. Move that 8 one column right to column 2, replacing the old position with zero: the **output** becomes [[0, 8]]. Pooling therefore does not guarantee that a classifier ignores every small movement.

Negative values are valid **input**s. For [[-4, -2], [-7, -3]], the maximum is -2 and the mean is -4. Starting a maximum calculation at zero would incorrectly return 0.

Adding a zero to the window would also change its maximum: padding choices matter. Our examples use no padding.

Returning the position of a maximum can preserve extra location information, but it still does not restore the discarded values.

For example, [[5, 1], [2, 3]] and [[5, 0], [0, 0]] have the same unique maximum 5 at the top-left position. Even the maximum plus its position cannot tell which original window was used.

Complete shift counterexample with an explicit head

Complete2×4 movement probe, all other values 0	Max map	Mean map	Declared probe head:class 1if left max>0
8 atrow0, column 1	[[8, 0]]	[[2, 0]]	1
Move8 torow0, column 2 andzeroitsoldposition	[[0, 8]]	[[0, 2]]	0

Complete2×4 movement probe, all other values 0	Max map	Mean map	Declared probe head:class 1if left max>0
What follows	Crossing the window boundary changes both maps	Not universal one-pixel invariance	This explicitly supplied head changes class; unknown other heads need their own checks

Try the ignored99border case

- Apply the lesson complete-window rule to [[1, 2, 99], [3, 4, 99], [99, 99, 99]]. Give the **output** shape and both **output**s. Explain why neither **output** includes 99.

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise summaries and collision explanations with less help

> State which positions are in each window before selecting its values.

Current task and response

- For the lesson window starting at [2, 2], list the four values, sum, maximum and mean. State the complete pooled **output** shape.

Compare windows [[0, 0], [6, 0]] and [[0, 0], [0, 6]]. Compute both summaries. Can a deterministic head receiving only either scalar distinguish bottom-left from bottom-right?

Explain your prediction to a partner before comparing results.

Supported W2 record

Use workbook W2 for the connected example “Pooling and **spatial information**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for Pooling and **spatial information**. Proposed explanation: "**Max pooling** turns negative numbers into zero". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured location-loss reasoning

- Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Reserve the independent new negative-valued map

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Prove that location can be lost

Check your reasoning

Explain the picture

Run a complete pooling and information-loss audit

> All windows,negative examples,borders and movement rules are supplied.

Current collision task and original pool2 function

Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

```
def pool2(image):
    h, w = len(image), len(image[0])
    if h < 2 or w < 2:
        raise ValueError("Need a complete 2 by 2 window")
    maxima, means = [], []
    for r in range(0, h - 1, 2):
        max_row, mean_row = [], []
        for c in range(0, w - 1, 2):
            values = [image[r+i][c+j]
                      for i in range(2) for j in range(2)]
            max_row.append(max(values))
            mean_row.append(sum(values) / 4)
        maxima.append(max_row)
        means.append(mean_row)
    return maxima, means

x = [[1, 4, 2, 0], [3, 2, 5, 1],
     [0, 6, 1, 3], [2, 4, 2, 7]]
assert pool2(x) == ([[4, 5], [6, 7]],
                    [[2.5, 2], [3, 3.25]])
```

Complete runnable collision and boundary probe program

```
def pool2(image):
    h, w = len(image), len(image[0])
    if h < 2 or w < 2:
        raise ValueError("Need a complete 2 by 2 window")
    maxima, means = [], []
    for r in range(0, h - 1, 2):
        max_row, mean_row = [], []
        for c in range(0, w - 1, 2):
```

```
            values = [image[r+i][c+j]
                      for i in range(2) for j in range(2)]
            max_row.append(max(values))
            mean_row.append(sum(values) / 4)
        maxima.append(max_row)
        means.append(mean_row)
    return maxima, means

x = [[1, 4, 2, 0], [3, 2, 5, 1],
     [0, 6, 1, 3], [2, 4, 2, 7]]
assert pool2(x) == ([[4, 5], [6, 7]],
                    [[2.5, 2], [3, 3.25]])
# Complete collision, boundary and movement probes under the same rule.
left = [[4,0],[0,0]]
right = [[0,4],[0,0]]
print('different positions, same summaries', pool2(left), pool2(right))
print('all-negative window', pool2([[-4,-2],[-7,-3]]))
print('ignored incomplete boundary', pool2([[1,2,99],[3,4,99],[99,99,99]]))
before = [[0,8,0,0],[0,0,0,0]]
after = [[0,0,8,0],[0,0,0,0]]
pre_max, pre_mean = pool2(before)
post_max, post_mean = pool2(after)
print('cross boundary before max/mean', pre_max,pre_mean)
print('cross boundary after max/mean', post_max,post_mean)
# One explicitly constructed head proves a prediction need not be shift-invariant.
print('head class1 if left max>0:', int(pre_max[0][0]>0),int(post_max[0][0]>0))
def max_with_position(window):
    candidates = [(value, r, c) for r,row in enumerate(window) for c,value in
enumerate(row)]
    # Row-major first position wins a tie; these two probes have a unique maximum.
    value, r, c = max(candidates, key=lambda item: item[0])
    return value, (r,c)

print('max plus position still collides:',
      max_with_position([[5,1],[2,3]]),max_with_position([[5,0],[0,0]]))
# Valid nonempty rectangular numeric maps assumed; no padding or hidden ReLU.
```

Normal,collision,negative,missing and movement probes

Complete pooling probe	Expected result / action
Normal main 4×4 map	Max[[4, 5], [6, 7]], mean[[2.5, 2], [3, 3.25]]

Complete pooling probe	Expected result / action
Two one 4/three 0 windows with different positions	Both max 4/mean 1; either scalar alone loses the 4 position
All-negative[[-4, −2], [−7, −3]]	Max−2, mean−4; zero is not an input or the correct maximum
3×3 map[[1, 2, 99], [3, 4, 99], [99, 99, 99]]	Only complete top-left window:shape 1×1, max 4, mean 2.5; 99 border unread
Input only 1×1	ValueError:Need a complete 2by2 window
Missing/ragged numeric map	Outside nonempty rectangular numeric contract; review before pooling
8 crosses column 1→2in a2×4 zero map	Max[8, 0]→[0, 8], mean[2, 0]→[0, 2]
Unique max 5at top-left in[[5, 1], [2, 3]]and[[5, 0], [0, 0]]	Same max and its position, but different discarded values; even max-plus-position is not full reconstruction
Add ReLU or zero padding to negative inputs	A different operation/boundary rule, not an implicit part of this pool 2 contract

Predict,pool,compare and bound the claim

1. Predict all main summaries and the complete collision/boundary probes before running or tracing pool 2. Then use the standalone probe program to check all-negative values, the unread incomplete border and a one-column movement. The movement example supplies its own simple head so a changed prediction is an actual declared calculation, not an assumption about unknown classifiers.

   Record which detail is lost and which coarse positions remain. The code assumes valid rectangular numeric maps; it does not learn pooling weights or show that compression improves an actual task.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: pool a new map containing negative values

> Apply the same complete-window rule without adding an unrequested activation.

Complete independent map and tasks

Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by

zero is an error here.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Pool a new map independently

Attempt this without hints

Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the pooling rule and lost-information evidence together

> The project record should say which task distinctions remain available.

Project artifact and evidence table

Add a labelled pooling and **spatial information evidence** sheet to Image-pattern Network Investigation. Show how you can compute max and **mean pooling** under an explicit window rule.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and remaining limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one computed summary and one revealing collision

> Explain why a smaller map is not automatically a better representation.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Compute max and **mean pooling** under an explicit window rule

> **My deciding step, evidence and one remaining question**
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> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Max pooling	Replacing each window by its largest value
Mean pooling	Replacing each window by the arithmetic mean of its values
Spatial information	Information about where values occur and how positions relate
Pooling collision	Different inputs producing the same pooled values

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For the lesson window starting at [2, 2], list the four values, sum, maximum and mean. State the complete pooled **output** shape.
- Apply the lesson complete-window rule to [[1, 2, 99], [3, 4, 99], [99, 99, 99]]. Give the **output** shape and both **output**s. Explain why neither **output** includes 99.
- A 2 by 4 map is zero except for 8 at row 0 column 1. Move the 8 to column 2 and set its old position to zero. Give the max and mean maps before and after. What claim does this refute?

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent four-window negative-valued map for the first individual attempt.

Current explanation and response

- Pooling can reduce the amount of data **process**ed by later layers. Whether the lost information is acceptable depends on the task.

  A task about whether any large response exists in a region may benefit from a maximum; a task requiring exact positions may need the unpooled map or another representation. Do not assume smaller always means better.

- Compare pooling choices with matched data splits and a declared evaluation rule. Fit model parameters on **training** data and choose alternatives using validation data, leaving final tests reserved.

  Next week combines these components in a small image-model experiment.

- A 2 by 4 map is zero except for 8 at row 0 column 1. Move the 8 to column 2 and set its old position to zero. Give the max and mean maps before and after. What claim does this refute?

> **A teammate says a 2 by 2 max-pooled map lets them reconstruct the original 4 by 4 map exactly and guarantees unchanged predictions after any one-pixel shift. Use one collision and one boundary example to challenge those claims.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

A maximum does not identify its location

INPUT WINDOW ROWS	MAXIMUM	MEAN
[4 0] and [0 0]	4	1
[0 4] and [0 0]	4	1
[4 4] and [4 4]	4	4

The first two windows differ in the position of their nonzero value yet have the same maximum and mean. The third has the same maximum but a different mean. Each summary retains some distinctions and discards others.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Pool a small map and show the detail it cannot retain.

By the end, I can	How I will show it
Compute max and mean pooling under an explicit window rule	Explain the deciding step, show a trace or test, and state one limit.
Use collisions to explain lost spatial information	Explain the deciding step, show a trace or test, and state one limit.
Test boundary and movement claims with revealing examples	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Pool a small map and show the detail it cannot retain.
- Which labelled step matches this detail? Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.
- 1. Left window
- 2. Compare summaries
- 3. Right window
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare summaries Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2. Pooling compresses each local window into a summary.

Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Changed-case question and optional hint

- Changed-case practice: Swap 1 and 4 within the left window. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Both pooled maps remain unchanged.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Both pooled maps remain unchanged. Max and mean ignore the positions of values inside the same window.

Repair and retain the record

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from **prediction** error. Short practice: Why should the penalty value not be reported as validation error?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

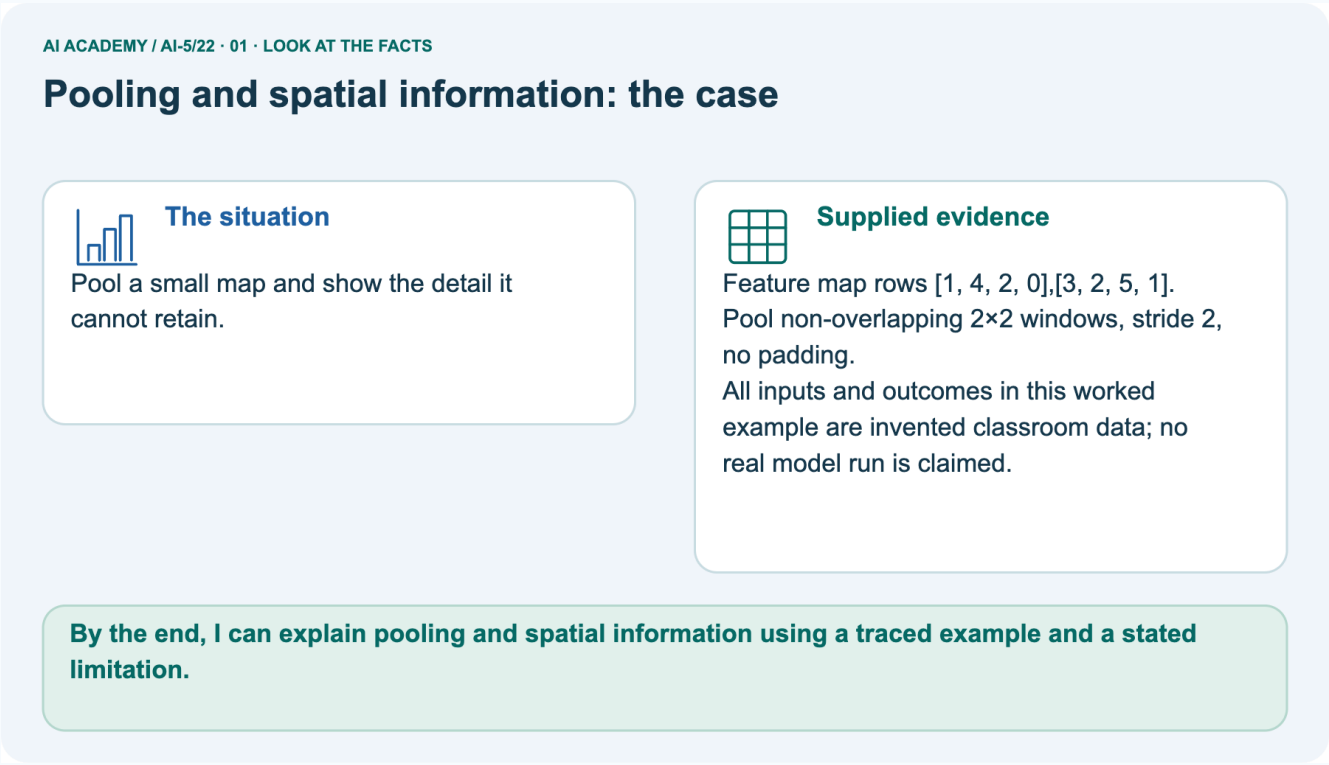
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Pooling and spatial information: the case



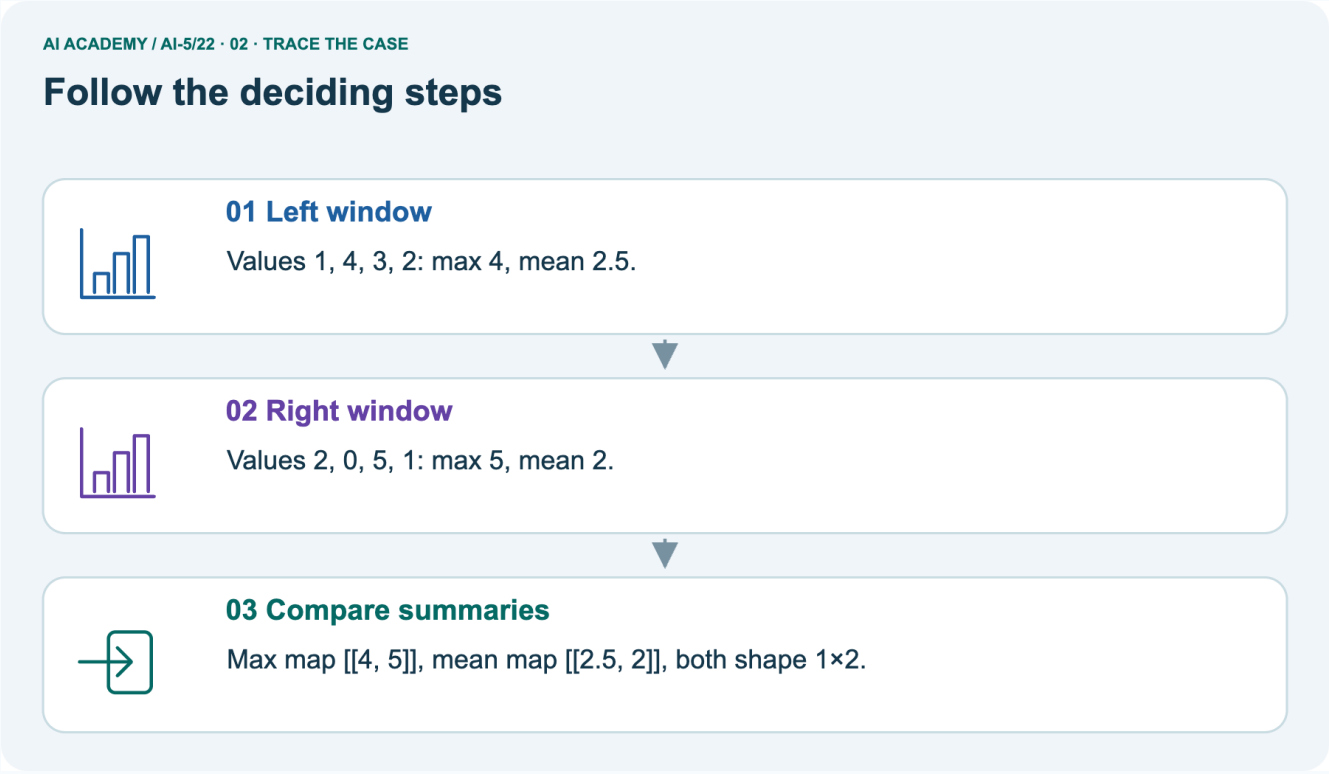
Follow the labelled picture with the matching explanation. Open the vector diagram

Pool a small map and show the detail it cannot retain. Supplied facts: Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Left window: Values 1, 4, 3, 2: max 4, mean 2.5. → Right window: Values 2, 0, 5, 1: max 5, mean 2. → Compare summaries: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two adjacent pooling windows

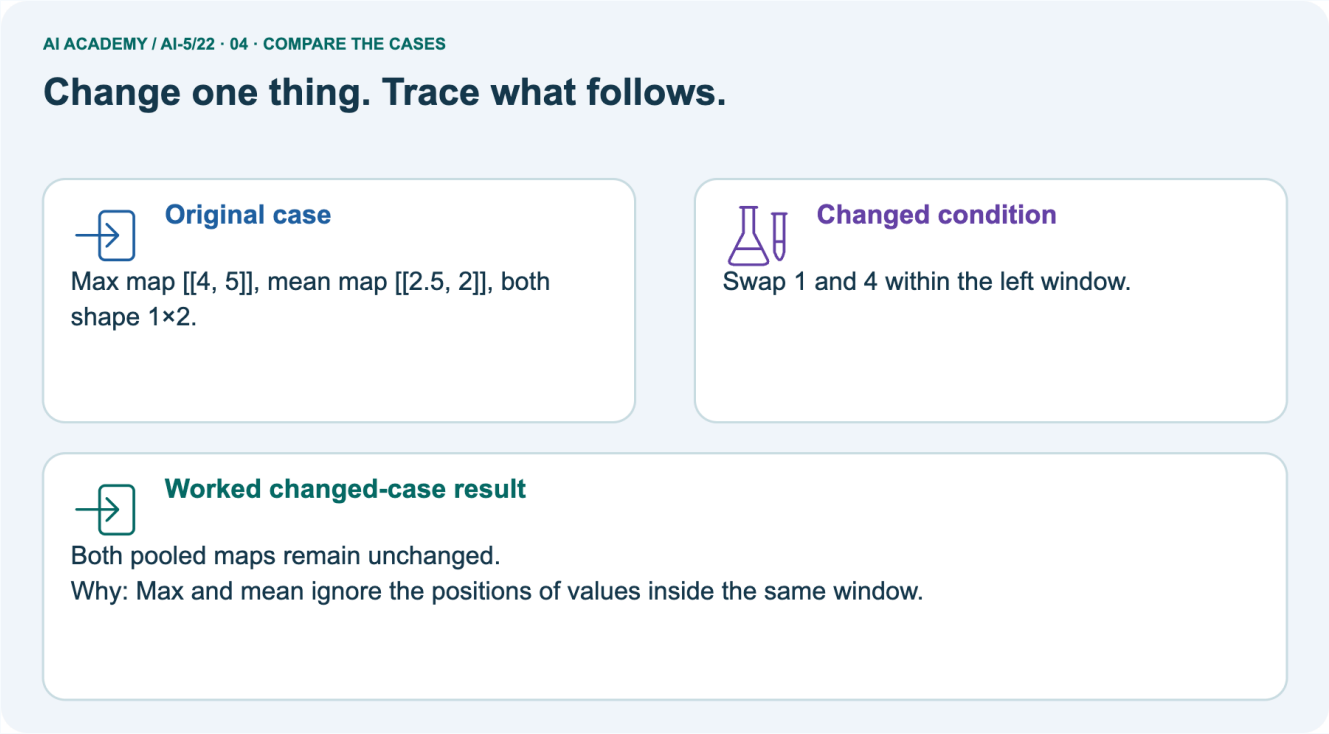


Follow the labelled picture with the matching explanation. Open the vector diagram

Two adjacent pooling windows. row 0: left 0 = 1, left 1 = 4, right 0 = 2, right 1 = 0. row 1: left 0 = 3, left 1 = 2, right 0 = 5, right 1 = 1.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



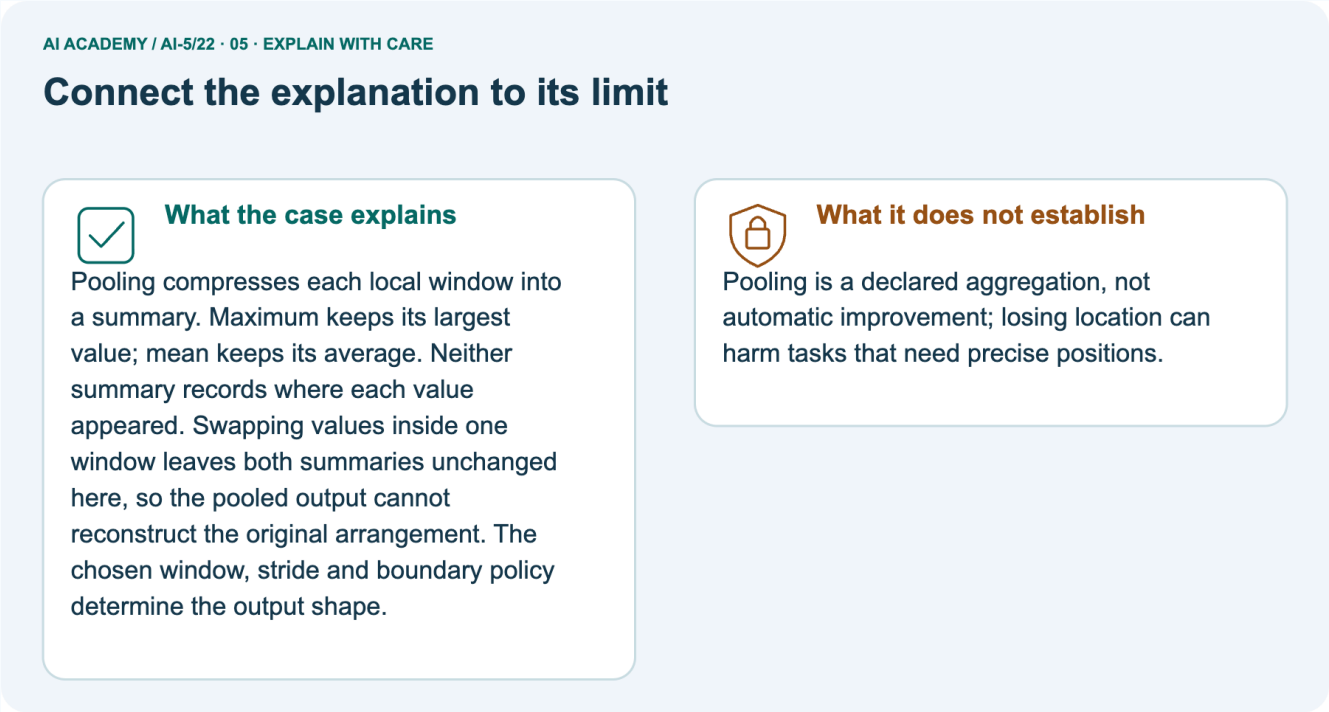
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2. Changed condition: Swap 1 and 4 within the left window. Expected result: Both pooled maps remain unchanged.

Reason: Max and mean ignore the positions of values inside the same window.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

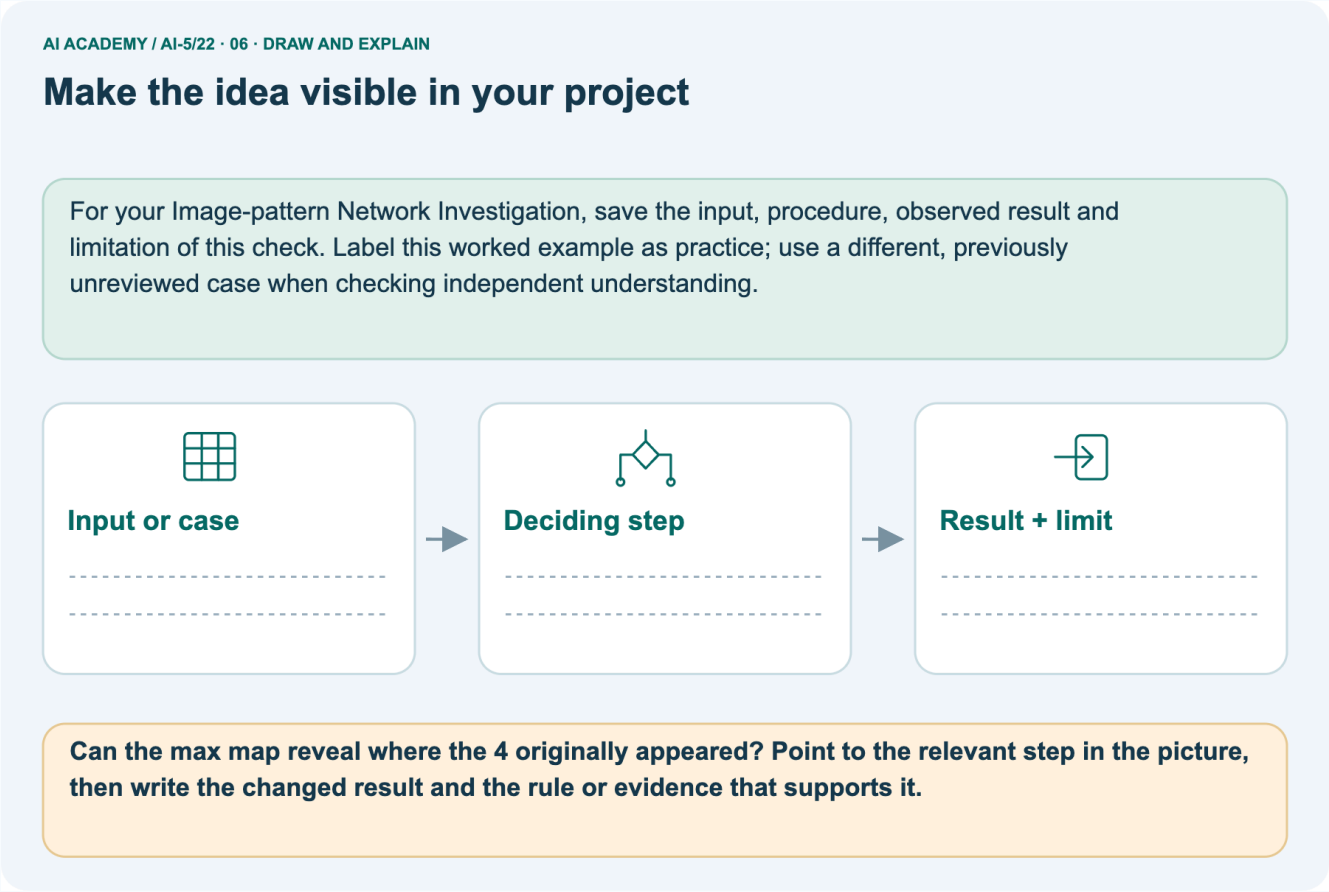
Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Limit: Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Pooling and spatial information: the case

02. Follow the deciding steps

03. Two adjacent pooling windows

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.